

Suvayu_Biswas

Full-Stack Developer | AI/ML | Cloud Certified

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PROFESSIONAL SUMMARY

Oracle and Red Hat Certified Computer Science student (**CGPA: 9.03**) with expertise in **Full-Stack Development, Backend System Design, and Applied AI/ML Research**. Built a **production-ready travel listing platform** and co-authored a **Springer-accepted publication** on lightweight deep learning for brain tumor classification. Seeking **Software Engineering Internship** opportunities to leverage strong problem-solving skills in **Java, JavaScript, Node.js, Python, and Cloud Technologies**.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science and Engineering,
Kalinga Institute of Industrial Technology (KIIT)
CGPA: 9.03 / 10

2023 – 2027
Odisha, India

PUBLICATIONS

Biswas, S., et al. “**Cascading Self-Attention Funnel (CSAF): A Lightweight Deep Architecture for Brain Tumor Classification from MRI**.” Accepted at **SCI-2026 International Conference**, Swinburne University, Hanoi, Vietnam. To appear in **Springer Lecture Notes in Networks and Systems (LNNS)**, 2026.

2026

PROJECTS & RESEARCH EXPERIENCE

Ghuumo – Full-Stack Travel & Listing Platform

09/2025 – 12/2025

GitHub : <https://github.com/SuvayuBiswas/ghuumo>

- **System Architecture:** Designed and developed a production-ready travel listing platform following the MVC architecture using **Node.js, Express.js, MongoDB, and EJS**.
- **Backend Development:** Built RESTful CRUD functionality for listings and reviews with **Mongoose** schemas, **Joi** validation, and middleware to ensure secure and reliable data management.
- **Authentication & Security:** Implemented secure authentication, authorization, and persistent user sessions using **Passport.js**, Express Sessions, input sanitization, and middleware-based access control.
- **Geospatial Integration:** Integrated **OpenStreetMap (Nominatim)**, **Leaflet.js**, and **Cloudinary** to provide geocoding, interactive maps, cloud-based image management, and location-based property discovery.
- **Database Design & UI/UX:** Engineered scalable MongoDB schemas with **Mongoose middleware** for cascading deletion of dependent reviews and developed a responsive **Bootstrap 5** interface with server-side rendering, category filtering, and location-based search.

Tech Stack: Node.js, Express.js, MongoDB, Mongoose, Passport.js, Joi, Leaflet.js, Cloudinary, Bootstrap 5

Cascading Self-Attention Funnel (CSAF) — Brain Tumor Classification Model,

06/2025 – 03/2026

Research Project | Accepted at SCI-2026 (Springer LNNS)

- **Model Innovation:** Architected a novel **lightweight deep learning model** utilizing a cascading self-attention mechanism to prioritize relevant tumor features while suppressing background noise.
- **Performance Optimization:** Achieved a **97.6% accuracy** and **99.67% recall** on the BRISC2025 dataset, outperforming standard heavy-weight CNN baselines (like ResNet) while using **<67K parameters**.
- **Resource Efficiency:** Optimized the model for deployment in resource-constrained environments (edge devices) by implementing aggressive **down-sampling** and attention funnels.
- **Research Impact:** The proposed CSAF architecture and experimental findings led to a peer-reviewed publication accepted at SCI-2026 (Springer LNNS).
- **Tech Stack:** Python, PyTorch, NumPy, Deep Learning, CNNs, Attention Mechanisms

Personal Developer Portfolio

12/2025 – 12/2025

GitHub : <https://github.com/SuvayuBiswas/suvayu.me>

- Designed and deployed a responsive portfolio to showcase projects, research, and skills.

TECHNICAL SKILLS

Programming Languages

JavaScript (ES6+), Python, Java, SQL, C

Frontend

HTML5, CSS3, Bootstrap, EJS, Responsive Design, DOM Manipulation, React.js

AI/ML

PyTorch, TensorFlow/Keras, CNNs, Attention Mechanisms, Data Preprocessing

Backend & Database

Node.js, Express.js, MongoDB, Mongoose, RESTful APIs

Core CS Concepts

OOP, DSA, API Integration, Data Validation, System Design

Cloud & DevOps

Oracle Cloud (OCI), AWS, Linux/Bash (RHEL), Git, GitHub

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 - Certified Architect Associate
- Oracle Cloud Infrastructure 2025 - Certified Generative AI Professional
- AWS Academy Graduate - Cloud Foundations, Amazon Web Services
- Red Hat System Administration I (RH124 - RHA) - Ver. 9.3
- JavaScript Algorithms and Data Structures - freeCodeCamp
- Responsive Web Design - freeCodeCamp